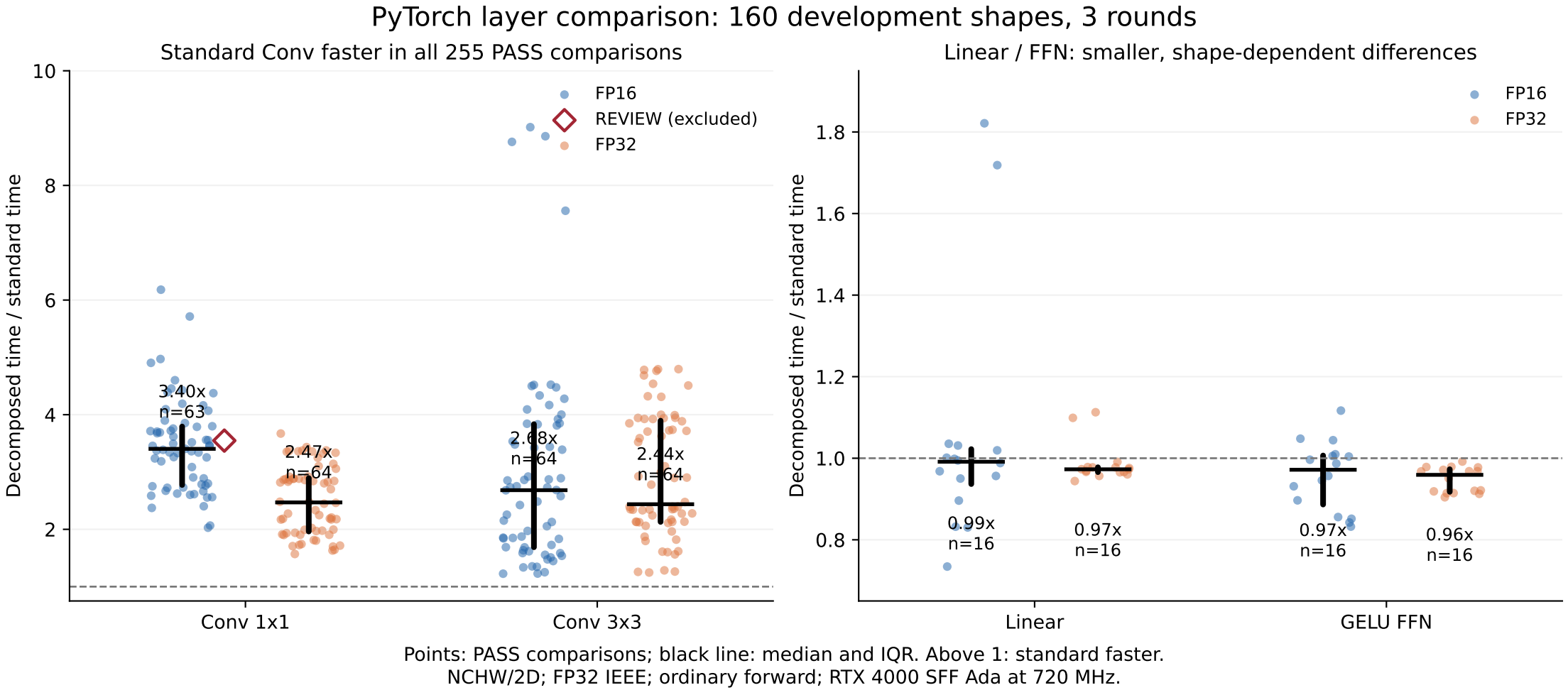
PyTorch layer comparison: 160 development shapes, 3 rounds
Standard Conv faster in all 255 PASS comparisons
Linear / FFN: smaller, shape-dependent differences
Decomposed time / standard time
FP16
REVIEW (excluded)
FP32
3.40x
n=63
2.47x
n=64
2.68x
n=64
2.44x
n=64
0.99x
n=16
0.97x
n=16
0.97x
n=16
0.96x
n=16
Conv 1x1
Conv 3x3
Linear
GELU FFN
Points: PASS comparisons; black line: median and IQR. Above 1: standard faster.
NCHW/2D; FP32 IEEE; ordinary forward; RTX 4000 SFF Ada at 720 MHz.